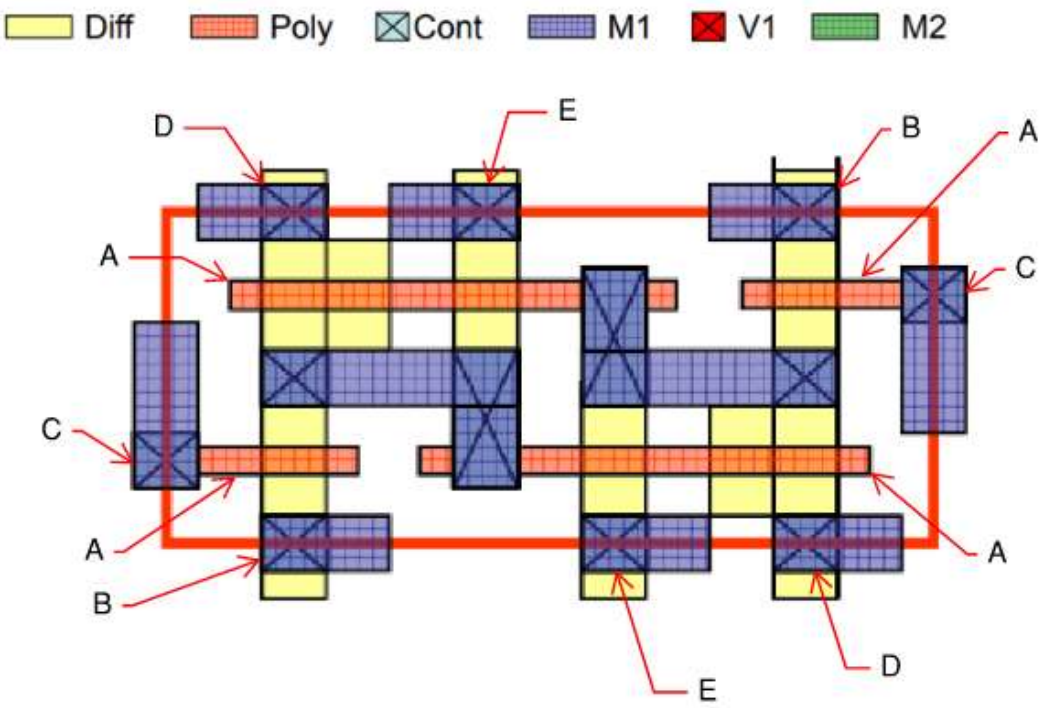
Q1:



(F) Bit Lines run horizontally in the metal one layer.
(G) Word Lines run vertically in the metal one layer.
(H) Again, Bit Lines run in horizontal tracks on metal one.
(F2) VSS & VCC run horizontally on metal one.

Q2:

2a) SRAM block capacity: 2^{12} = 4096 (64-bit) words

4096 * 64 = 262,144 bits (256kb)

2b) Standard Bit Cell: Width = 0.2um, Height = 0.4 um

262,144 * 0.2um * 0.4um = 20971.52 um^2

For a square we want array width = array height

So bits per row * width = rows * height

64x * 0.2 = 4096/x * 0.4 -> 12.8x = 1638.4/x -> 12.8x^2 = 1638.4 -> x ~= 11.3, so we choose 8 words a row since a mux requires a power of 2 for inputs and outputs.

4096 bits/8 words = 512 rows

Bits per row = 8 * 64 = 512 bits

Width = 512 * 0.2 = 102.4 um

Height = 512 * 0.4um = 204.8

Row decoder: $\log_2(512) = 9$, so 9 inputs, 512 outputs

Column decoder: $\log_2(8) = 3$, so 3 inputs, 8 outputs

2c) Total bits: 262,144

1nA leakage: 10^{-9} * 262,144 = 262.144 uA

At 1.1V: P_{leak} 262.144 uA * 1.1V = 288.4 uW

2d) 9 * 2^20 = 9 * 1,048,576 bytes = 75,497,472 bits

1nA * 1.1V = 1.1nW

1.1nW * 75,497,472 bits = 83.05mW

2e) $T_{ox} = 1.5nm$

$f = 10^{-15}$

$C_{ox} = E_{ox} / t_{ox} = (3.9 * 8.854 * 10^{-12}) / (1.5 * 10^{-9}) \sim= 23fF/um^2$

$C_{ox} * W * L = 0.46fF$

$2.0\ fF/um * 0.2um = 0.4fF$

$C_{gate} = 0.46fF + 0.4fF = 0.86fF$

$C_{junction} = 0.5fF$

$C_{WL} = 64 * 2 * 0.86fF = 110.08fF$

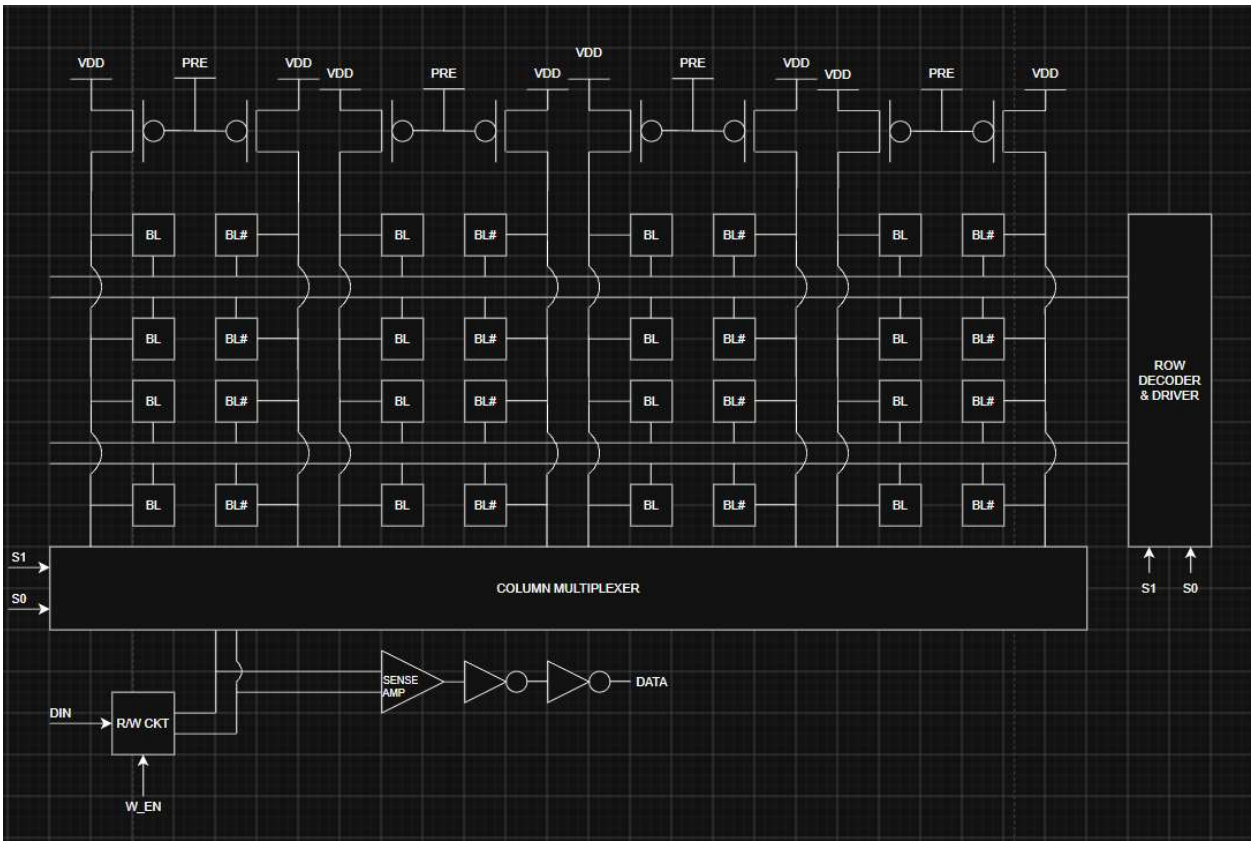
$C_{BL} = 64*.5fF = 32fF * 2 = 64fF$

$C_{total} = 110+64 \sim= 174fF$

$P_{cap} = CV^2 * f = 174 * 10^{-15} * (1.1)^2 * 2 * 10^9 = 0.42108\ mW$

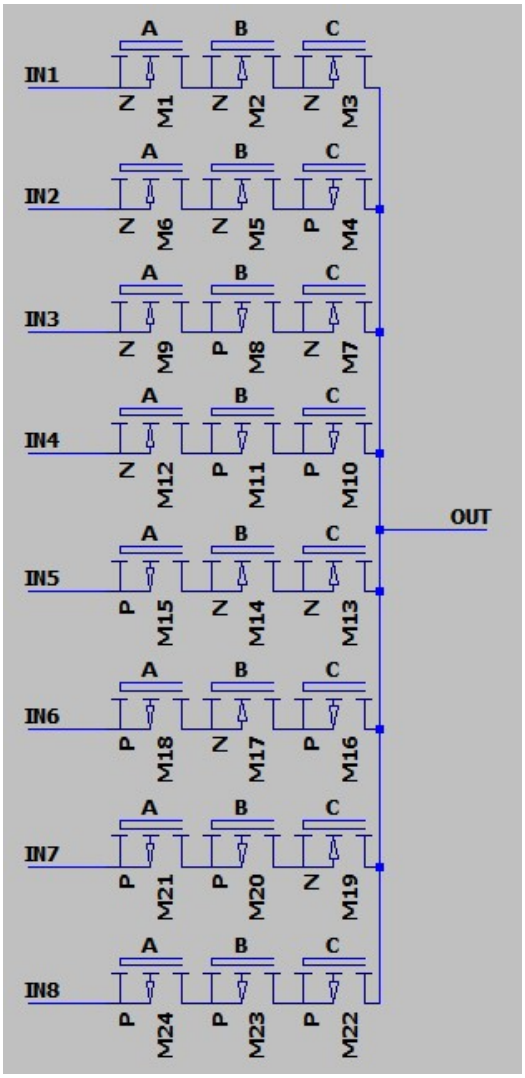
$P_{total} = P_{leak} + P_{cap} = 0.42108 + 0.2884 = 0.70948\ mW$

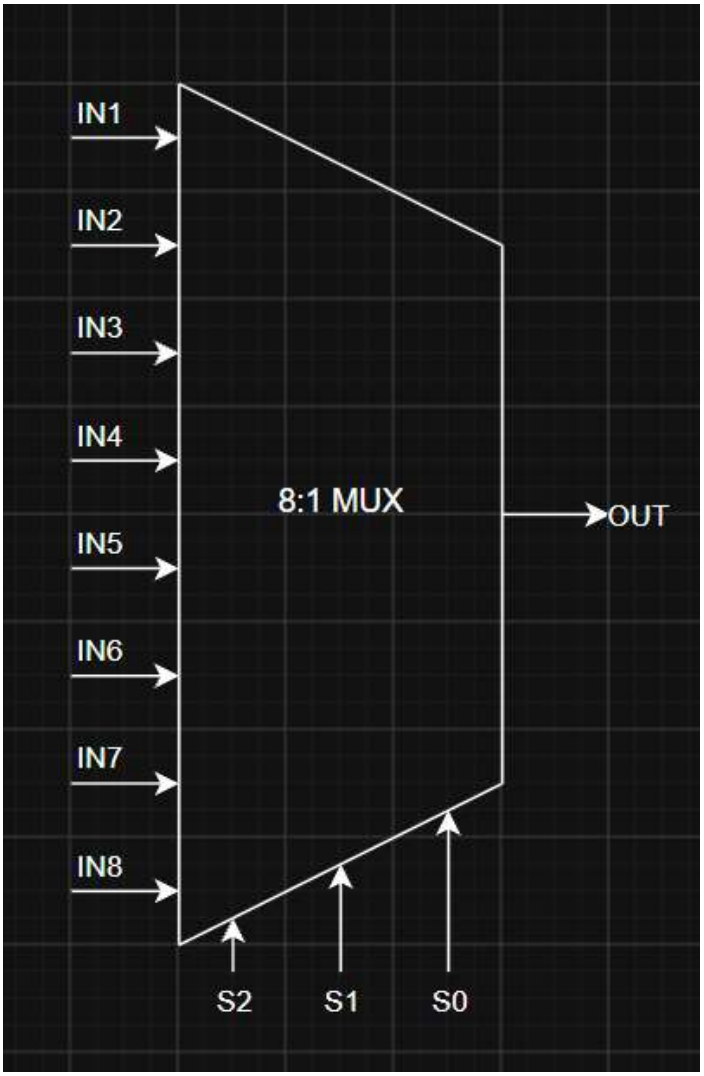
Q3:



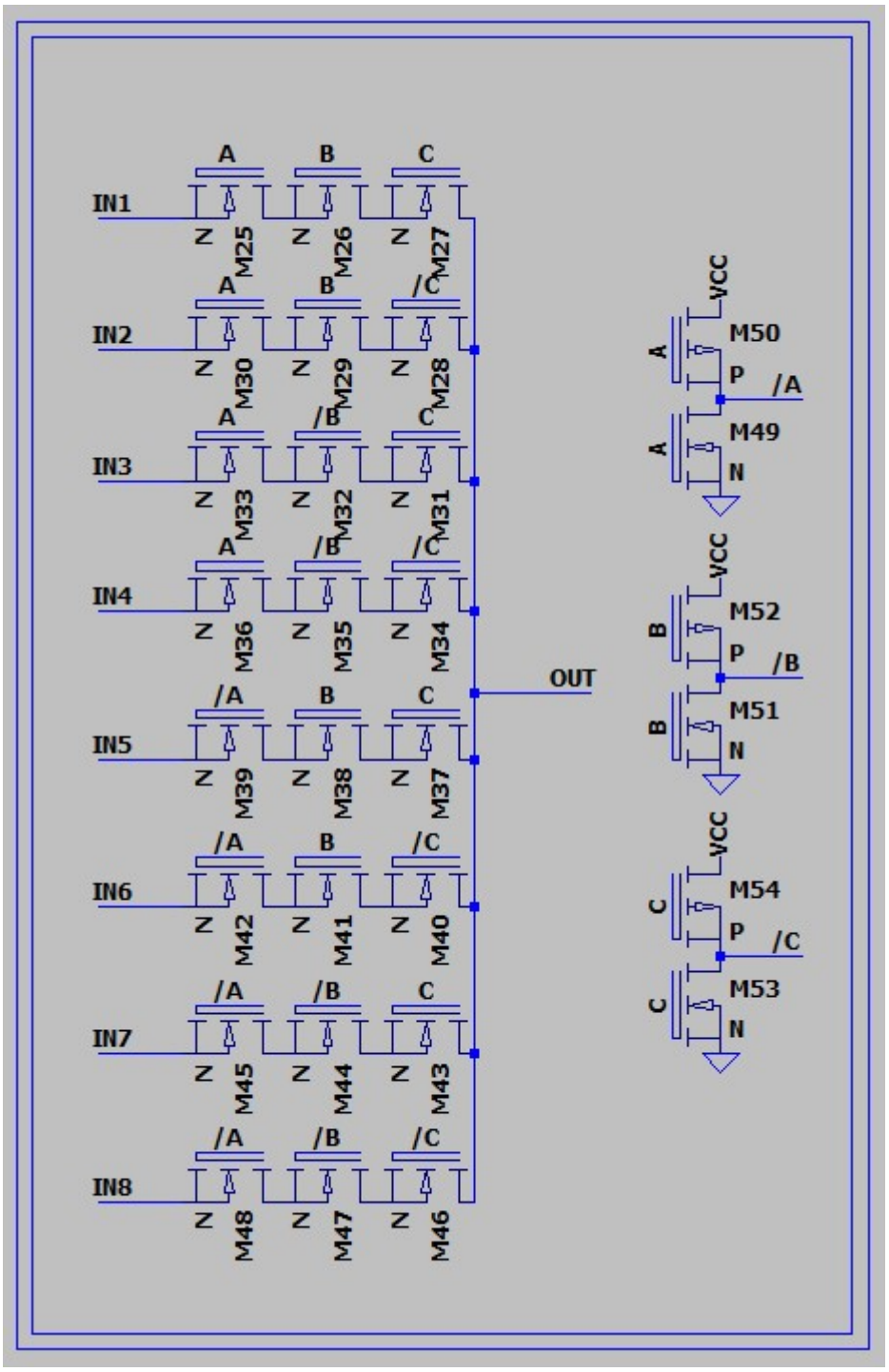
Q4: The gate is a NAND, with two NMOS in series, when both are true it pulls the output to ground upon a rising edge clock, otherwise outputting Vdd.

Q5: 8-1 Multiplexer Schematic (Left), Block (Right)



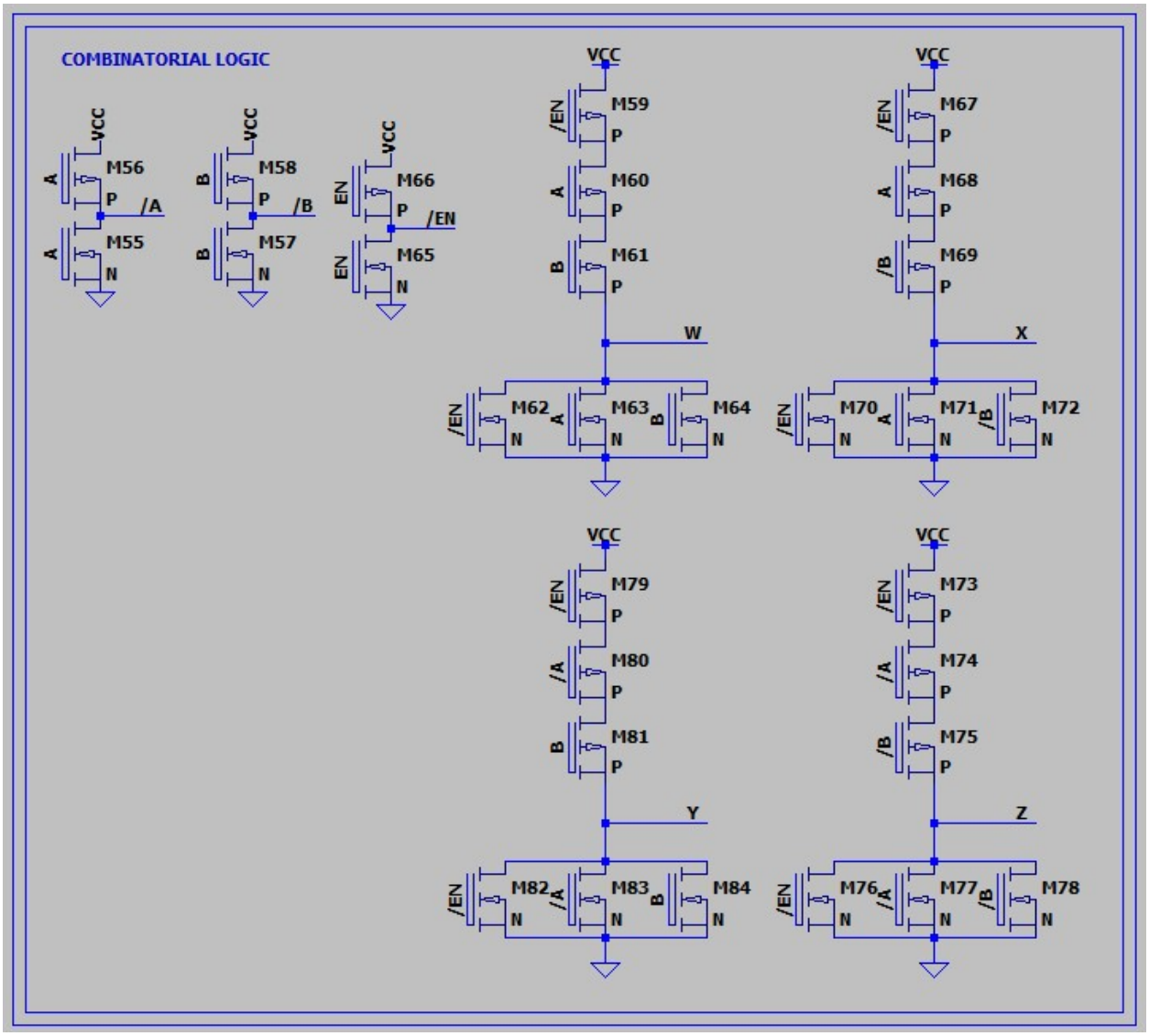


Q6: Used only NFET for the mux design, assumed we can use PFET for inverters per standard CMOS design.



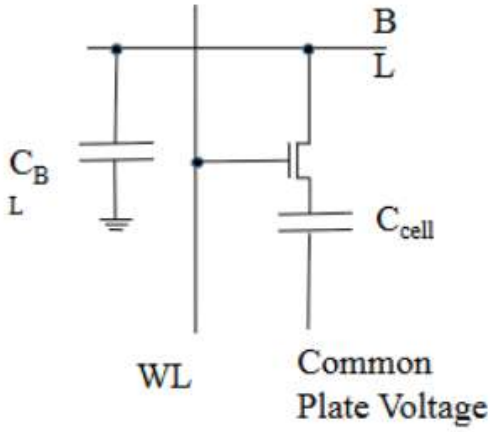
Lower noise margin. Since it does not pull as high (VDD – VT) there is less range for it to fluctuate before it could accidentally read as a 0 instead of 1.

Q7:


COMBINATORIAL LOGIC
VCC
M56
M55
M58
M57
M66
M65
/A
/B
/EN
M59
M60
M61
M62
M63
M64
W
M67
M68
M69
M70
M71
M72
X
M79
M80
M81
M82
M83
M84
Y
M73
M74
M75
M76
M77
M78
Z

Q8:

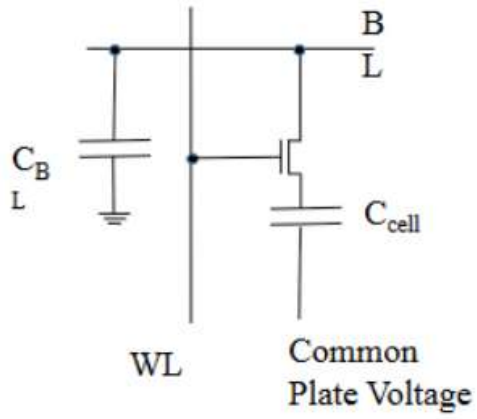
Problem 8: Assume a bit line capacitance of 1pF, a precharge voltage of 1.25V, cell capacitance of 50fF, V_{TN}=0.5V, 2.5V = stored logic 1, 0V = stored 0. What is the magnitude of voltage swing on the bit line during a read operation.



Swing $= \frac{C_{cell}(V_{cell}-V_{pre})}{C_{BL}+C_{cell}} = 50fF * \frac{2.5-1.25}{1000fF+50fF} = 50 * \frac{1.25}{1050} \cong 59.5mV$

Q9:

Problem 9: Assume a bit line capacitance of 1pF, a precharge voltage of 1.25V, cell capacitance of 50fF, V_{TN}=0.5V, 2.5V = stored logic 1, 0V = stored 0. If the leakage (I_{OFF}) through the pass transistor is 1fA, estimate the required DRAM refresh time (assume refresh is required when 10% of the storage electrons leak from the storage node)



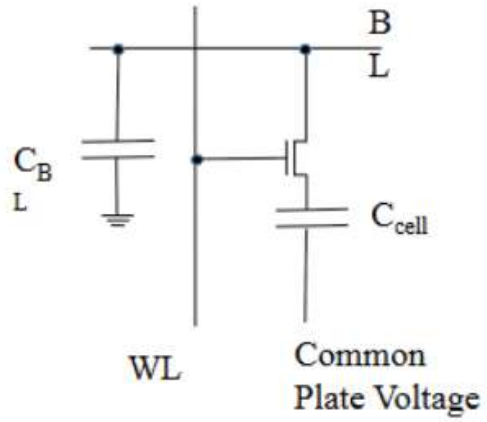
To find charge we do Q = CV, or $Q = C_{cell}V = 50fF * 2.5V = 125fC$

.1Q = 12.5fC

T = Q/t = 12.5f/1fA = 12.5s refresh time.

Q10:

Problem 10: Assume a bit line capacitance of 1pF, a precharge voltage of 1.25V, cell capacitance of 50fF, V_{TN}=0.5V, 2.5V = stored logic 1, 0V = stored 0. IF 2.5V is applied to the WL and BL during a write operation, what is the voltage level for a Logic 1. How could the write operation be changed to get 2.5V on the storage node for a logic 1?



$V_{cell} = V_{WL} - V_{TN} = 2.5 - 0.5 = 2.0$, to get it to 2.5V on the output we would want to raise our V_{WL} to 3.0V, so that 3.0 – 0.5 = 2.5V.